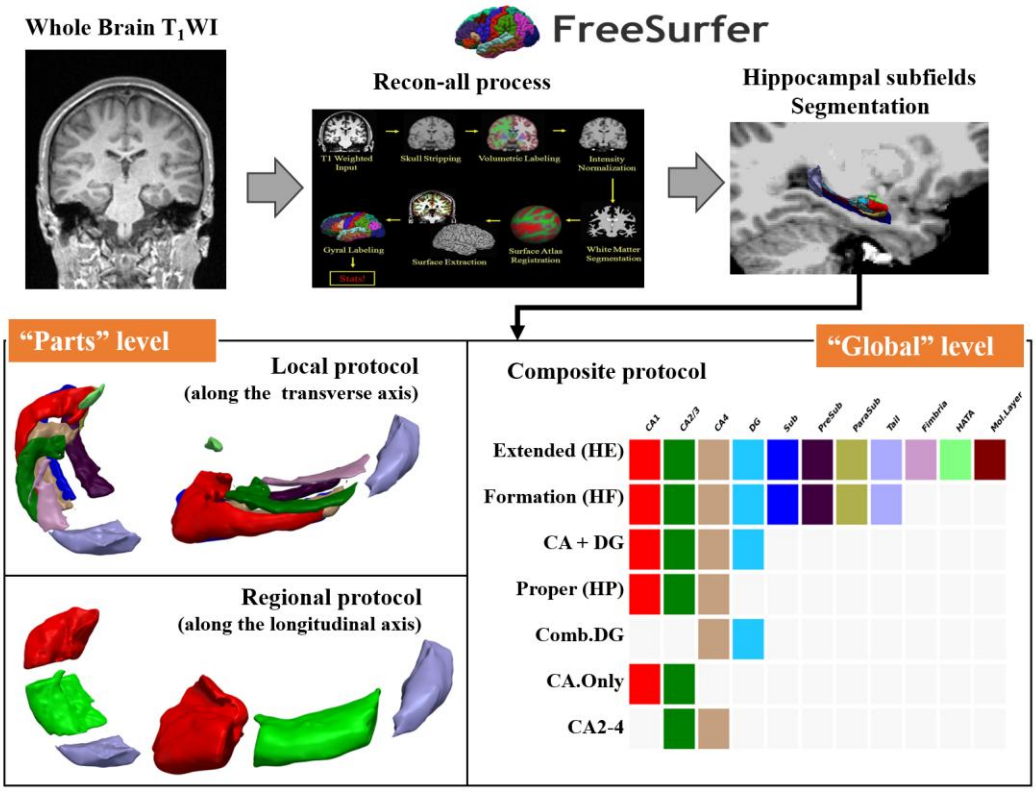


**Figure 1.** Image processing pipeline and hierarchical hippocampal segmentation strategies. Whole-brain T1-weighted images were processed using the FreeSurfer (v7.1.1) recon-all pipeline and the hippocampus were further segmented into subfields. Final metrics used for later analyses followed three protocol: (1) The Composite Protocol ("Global" Level, bottom right): various existing "global hippocampus" definitions that includes different set of subfields; at the "parts" level, (2) The regional protocol (left bottom), macro-scale segmentation along the longitudinal axis into head, body, and tail; (3) the local protocol (left top),micro-scale segmentation along the transverse axis into discrete histological subfields. Abbreviations: CA, Cornu Ammonis; DG, Dentate Gyrus; Sub, Subiculum; PreSub, Presubiculum; ParaSub, Parasubiculum; HATA, Hippocampal-Amygdaloid Transition Area; Mol. Layer, Molecular Layer; HE, Hippocampal Extended; HF, Hippocampal Formation; HP, Hippocampal Proper; Comb.DG, Combined Dentate.

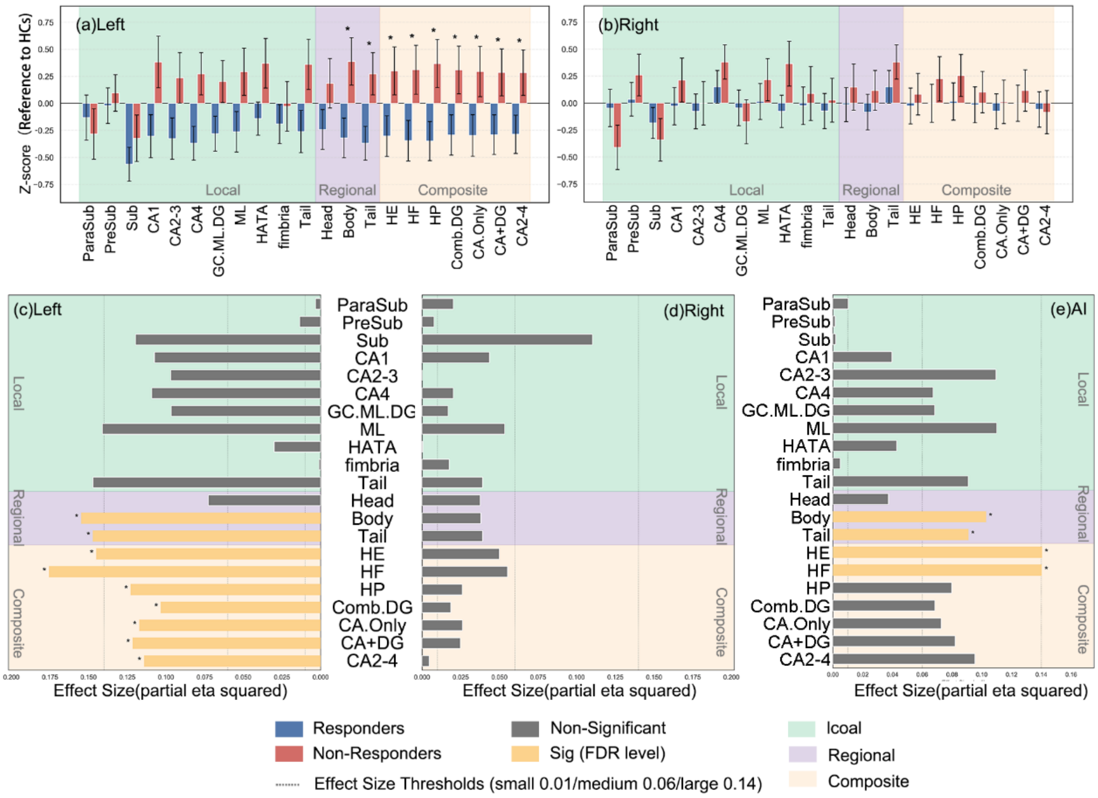


**Figure 2. Volumetric profiles and effect sizes distinguishing Responders from Non-responders across segmentation protocols.**

**(a, b)** Z-score standardized volumes relative to the healthy control baseline for the left **(a)** and right **(b)** hemispheres. Blue bars represent responders; red bars represent non-responders. Error bars indicate 95% confidence intervals. Effect sizes (partial eta squared) for group differences in volumes **(c)** and asymmetry Index **(d).** Vertical dotted lines represent thresholds for small (0.01), medium (0.06), and large (0.14) effect sizes. Asterisks denote FDR level significance within protocols.

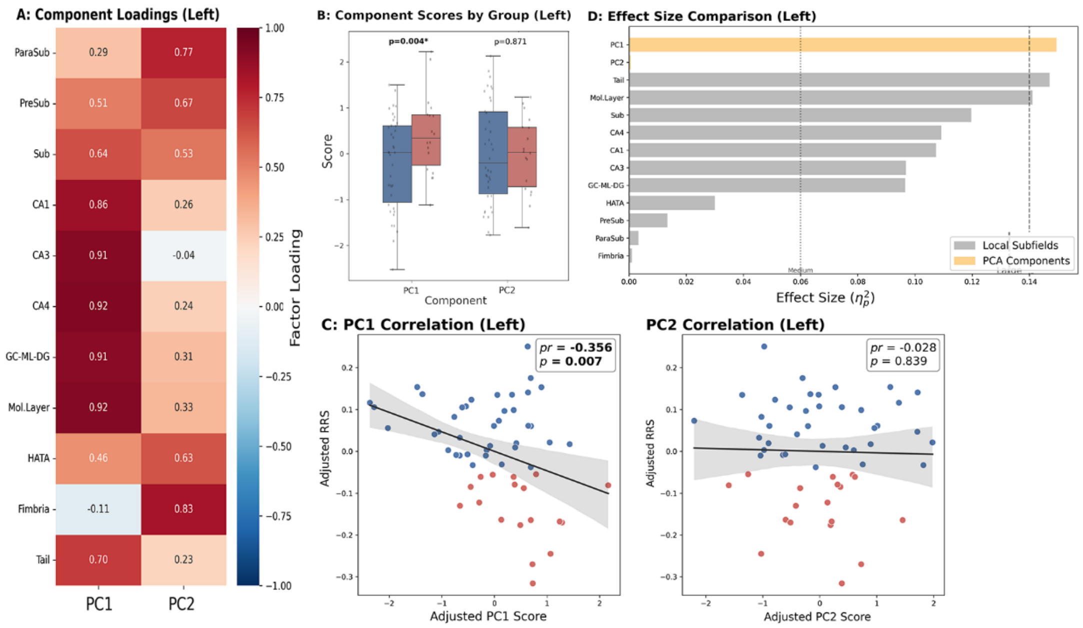


Figure 3. Data-driven identification of a latent "core hippocampal factor" in the local protocol, left hemisphere. (A) Heatmap displaying the contribution of each local subfield to the two extracted principal components (PC1 and PC2). PC1 is heavily loaded by the core subfields (CA1-4, Mol. Layer, DG), representing a shared anatomical variance. This PC1 score showed significant difference between responders and non-responders (B), and significant correlation with HAMD reduction rate (C). Effect size comparison showed the effect size (partial eta squared) of the latent PC1 factor surpasses all single subfields(D). Dotted line indicates Large Effect Size threshold (0.14).

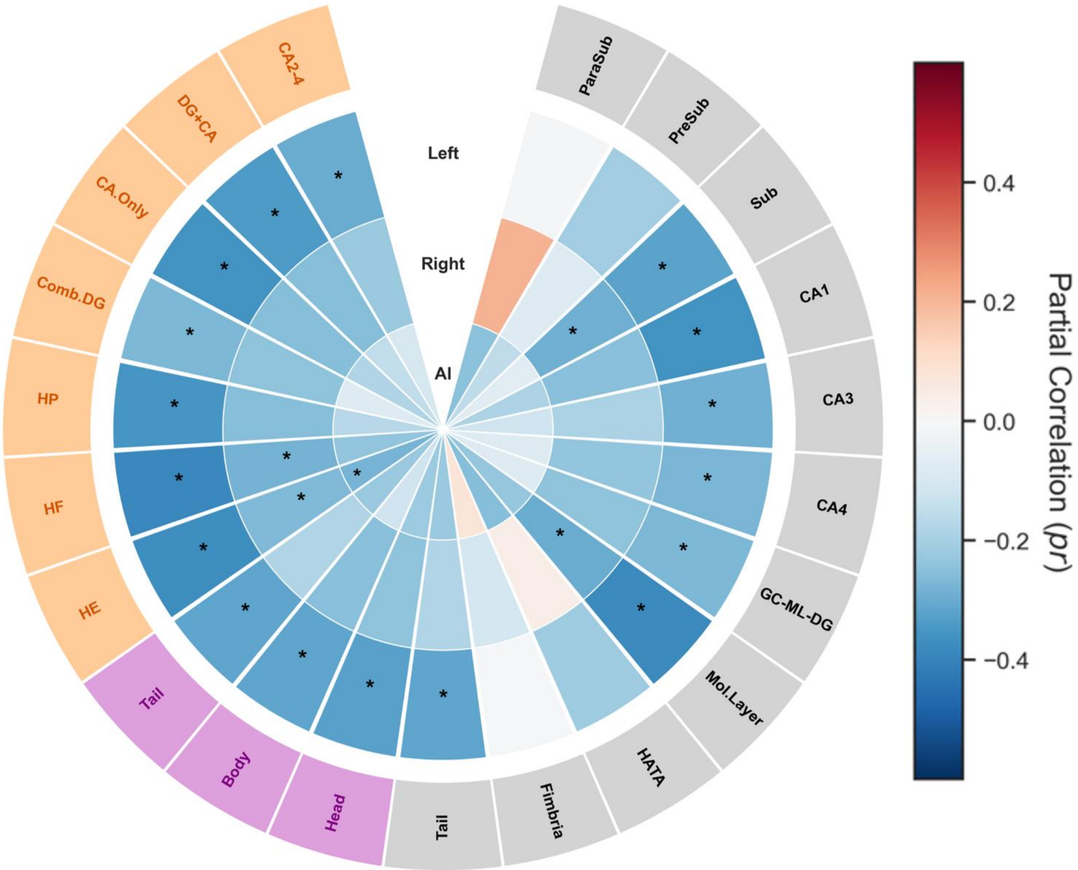


**Figure 4. Circular heatmap showing the association between hippocampal metrics and antidepressant response.**

The heatmap displays partial correlation coefficients ($pr$) between the HAMD reduction rate and hippocampal metrics, adjusted for age, sex, and ICV. The plot is organized into concentric rings representing the left hemisphere (outer), right hemisphere (middle), and asymmetry index (inner). Segments are grouped by segmentation protocol: composite (orange labels), regional (purple labels), and local (gray labels). Asterisks (*) denote nominal statistical significance ($P<0.05$, uncorrected).

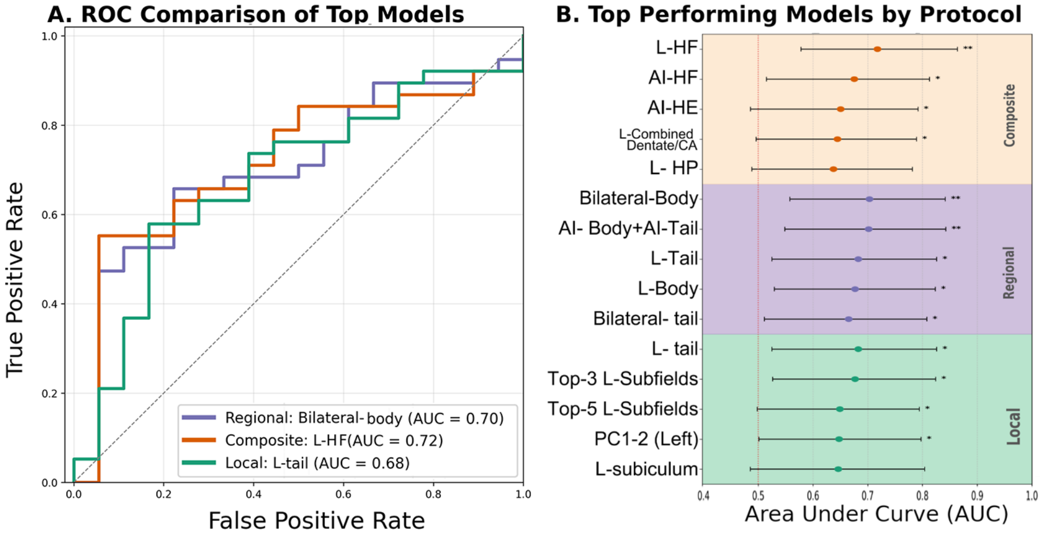


**Figure 5. Predictive performance of top models across segmentation protocols.**

**(a)** ROC curves for the best-performing feature set within each protocol.

**(b)** AUC values for the top 5 models in composite (orange background), regional (purple), and local (green) protocols. Error bars represent 95% confidence intervals derived from bootstrapping. Asterisks indicate statistical significance based on permutation testing (* $p < 0.05$, ** $p < 0.01$). Vertical dotted line represents random chance (AUC = 0.50).